

Regularization and Penalized Regression

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Consider the basic linear regression model:

$$Y = w_0 + w_1 X_1 + w_2 X_2 + \dots + w_p X_p + \epsilon$$

We've previously estimated $\mathbf{w}$, the vector of weights, by optimizing the following cost function:

$$\text{Cost} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{1}{n} (\mathbf{y} - \mathbf{X}\hat{\mathbf{w}})^T (\mathbf{y} - \mathbf{X}\hat{\mathbf{w}})$$

Regularized Regression

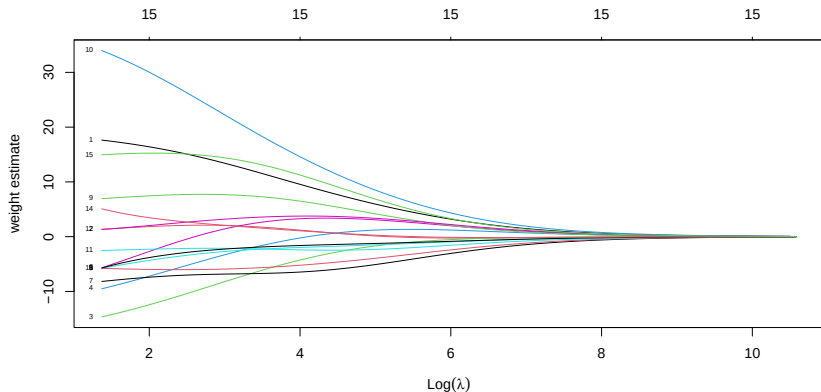
Regularized regression adds a penalty term to the cost function that shrinks weight estimates towards zero:

$$\text{Cost} = \frac{1}{n}(\mathbf{y} - \mathbf{X}\hat{\mathbf{w}})^T(\mathbf{y} - \mathbf{X}\hat{\mathbf{w}}) + P_{\alpha}(\hat{\mathbf{w}})$$

- ▶ $P()$ is a *penalty function* involving α , a **regularization parameter** that controls the trade-off between each term in the cost function

Example

When the regularization parameter, α , is large, the penalty term dominates the cost function and weights are estimated to be zero. When α is zero, cost function reduces to squared error loss.

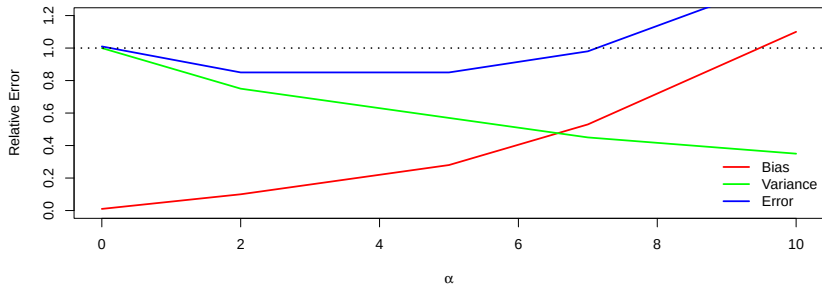


Benefits of Regularization

- ▶ Intuitively, the premise behind regularization is that small weights should occur more frequently than large weights when many predictors are considered
 - ▶ Thus, using penalization to discourage larger estimated weights can prevent overfitting
- ▶ In 1970, Hoerl and Kennard proved that *ridge regression* (a type of regularized regression) can *always* produce a lower out-of-sample *RMSE* than ordinary (unpenalized) regression

Benefits of Regularization

Mathematically, it's possible to decompose mean-squared error (MSE) into bias and variance terms. Here's a heuristic look at how these components might look as α is varied:



Ridge Regression

Ridge regression uses the penalty function: $P_{\alpha}(\mathbf{w}) = \alpha \sum_{j=1}^p w_j^2$

In matrix form, the Ridge regression cost function looks like:

$$\text{Cost} = \frac{1}{n}(\mathbf{y} - \mathbf{X}\hat{\mathbf{w}})^T(\mathbf{y} - \mathbf{X}\hat{\mathbf{w}}) + \alpha \hat{\mathbf{w}}^T \hat{\mathbf{w}}$$

- ▶ $\hat{\mathbf{w}}^T \hat{\mathbf{w}}$ is the squared *L2 Norm* of the weight vector (or $\|\hat{\mathbf{w}}\|_2^2$), so the ridge penalty is often called *L2 regularization*
- ▶ The meaning of α is entirely relative, so sometimes you'll see the cost written using $P_{\alpha}(\mathbf{w}) = \frac{1}{2n} \alpha \sum_{j=1}^p w_j^2$

Ridge Regression

Similar to ordinary linear regression, minimizing the ridge regression cost function has a closed-form solution:

$$\hat{\mathbf{w}} = (\mathbf{X}^T \mathbf{X} + \alpha \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y}$$

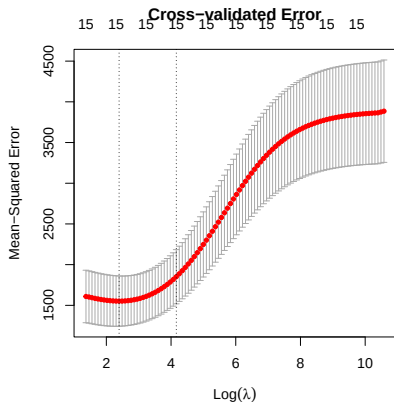
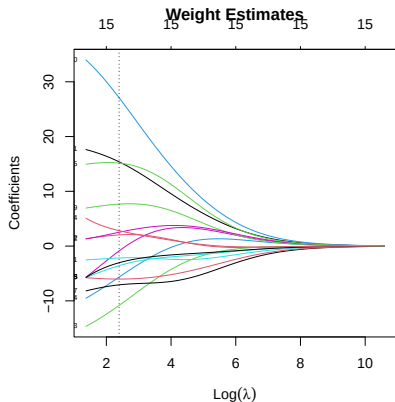
The method gets its name from the “ridge” added to the diagonal of $\mathbf{X}^T \mathbf{X}$ prior to inversion

Choosing α

- ▶ In penalized regression, α is a tuning parameter, with different values leading to different weight estimates
 - ▶ Larger values of α shrink the weights closer to zero (introducing more bias while reducing variance)
 - ▶ When $\alpha = 0$, the ridge regression estimates are the same those of ordinary linear regression
- ▶ Because penalization is proportional to the magnitude of w_j , it is important to *standardize* each variable as a pre-processing step when using regularization

Choosing α (example)

Below are results for data that uses pollution and demographic variables of 60 US metro areas to predict age-adjusted mortality:



- ▶ The ridge penalty provides *stability* (ie: reduces variance) at the expense of adding *bias*
 - ▶ However, it doesn't truly reduce the complexity of the model (the number of non-zero weights is the same, regardless of the amount of penalization)

- ▶ The ridge penalty provides *stability* (ie: reduces variance) at the expense of adding *bias*
 - ▶ However, it doesn't truly reduce the complexity of the model (the number of non-zero weights is the same, regardless of the amount of penalization)
- ▶ The lasso (least absolute shrinkage and selection operator) addresses this shortcoming by promoting *sparsity* in the estimated weight vector
 - ▶ The lasso penalty function is: $P_{\alpha}(\mathbf{w}) = \alpha \sum_{j=1}^p |w_j|$
- ▶ Recognize that the absolute value function is not strictly differentiable at its minimum
 - ▶ This promotes weight estimates of exactly zero (sparsity)

- ▶ To better understand why the lasso penalty promotes sparse weight estimates, we can view minimizing the lasso cost function as a constrained optimization problem
 - ▶ That is, the lasso's estimate of $\mathbf{w}$ minimizes $\frac{1}{n} \sum_{i=1}^n (y_i - \mathbf{x}_i^T \mathbf{w})^2$ subject to the constraint $\sum_{j=1}^p |w_j| < c$ where c describes a fixed amount of penalization (a function of α)
 - ▶ For comparison, the ridge estimate is similar but with the constraint $\sum_{j=1}^p w_j^2 < c$
- ▶ The next slide provides a geometric illustration of why the lasso constraint promotes sparsity, but the ridge constraint does not

Lasso vs. Ridge

Estimates satisfying $\sum_{j=1}^p |w_j| < c$ exist within a diamond, while those satisfying $\sum_{j=1}^p w_j^2 < c$ exist within an ellipse.

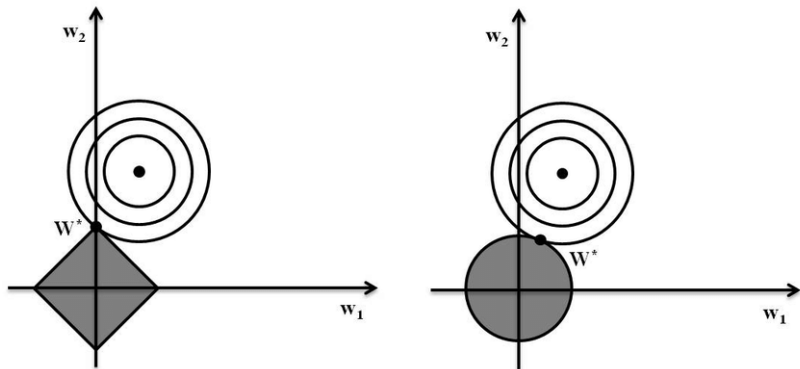
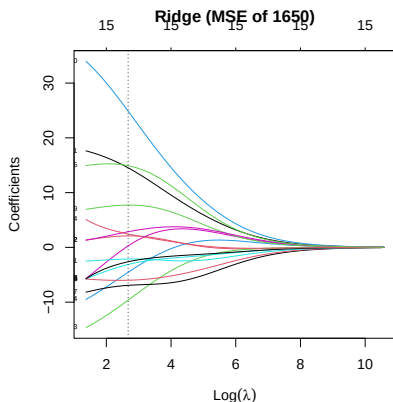
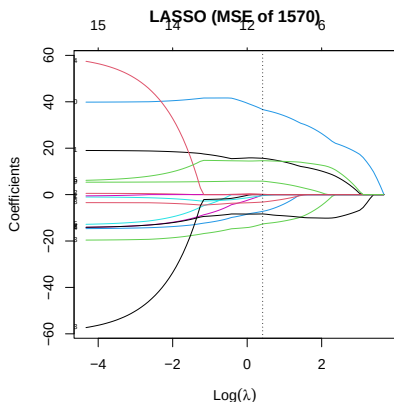


image credit: https://www.researchgate.net/figure/Plot-demonstrating-the-Sparsity-caused-by-the-LASSO-Penalty-The-plot-shows-the_fig1_317357840

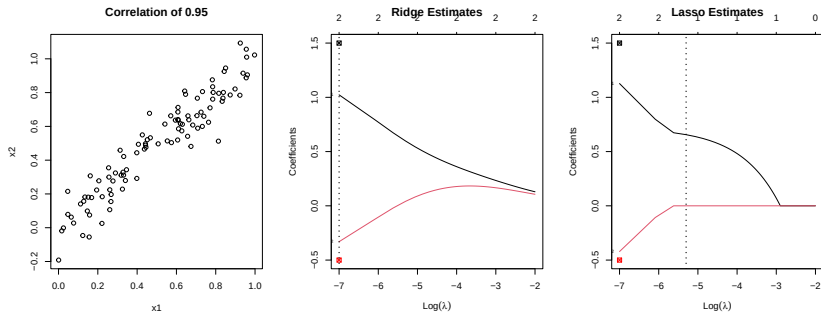
Lasso

For pollution example, lasso achieves a minimum cross-validated mean-squared error of around 1570, while ridge regression's minimum error (shown in an earlier slide) is around 1650 for these data.



Ridge Regression and Multicollinearity

- ▶ Consider data where $y_i = 1.5 * x_{i,1} - 0.75 * x_{i,2} + \epsilon$ where X_1 and X_2 have a correlation of 0.95
 - ▶ lasso favors a single representative, while ridge will distribute the weight estimates in a more balanced manner:



Final Remarks on Regularization

- ▶ L1 (lasso) and L2 (ridge) regularization can be used in many different machine learning models to help balance the bias-variance trade-off
- ▶ By default, the implementation of logistic regression in `sklearn` includes L2 regularization to promote stable weight estimates
 - ▶ In this regard, regularization can be used to address the “perfect separation” issue that arose in our previous lab